Predicting Breast Cancer Diagnosis Using Machine Learning

ISA 640: Capstone Project

Summer 2026

Project Milestone 3: Baseline Model

This milestone establishes the baseline model, the simplest reasonable solution to the task, and the benchmark every later model (tuned versions, ensembles) must beat to justify its added complexity.

Model Selection and Justification

The baseline model chosen is Logistic Regression. This choice reflects the structure of the data, the task, and practical constraints:

- Data type and structure: the 27-feature engineered set is dense, continuous, tabular data with no text, image, or time-series structure, the setting logistic regression is built for.
- Task type: binary classification (malignant = 1, benign = 0) maps directly onto logistic regression's modeling of the log-odds of the positive class.
- Interpretability: in a diagnostic-support context, being able to describe how a feature changes the odds of malignancy has real value to a clinical stakeholder, a property logistic regression's coefficients provide directly, without a separate explainability layer.
- Computational cost: with 569 rows, training and inference both complete in single-digit milliseconds, making it an efficient, iterable baseline before considering heavier models.

Conceptually, logistic regression models the log-odds of the positive class as a linear combination of the input features, then passes that linear score through the sigmoid function to produce a probability between 0 and 1. Training finds the coefficients that maximize the likelihood of the observed labels (equivalently, minimize log-loss); at prediction time, a 0.5 probability threshold converts that into a class label. Because its decision boundary is linear in the (scaled) feature space, it performs best when classes are close to linearly separable, which the Milestone 2 exploratory analysis suggested is roughly true here for the size/shape-irregularity features.

Handling of Class Imbalance in Modeling

The 1.68:1 imbalance identified in Milestone 2 was carried into the modeling split: an 80/20 stratified train/test split preserved the ratio in both sets (62.6%/37.4% in training, 63.2%/36.8% in test). No resampling was applied, so there is no risk of the common leakage mistake of resampling before splitting. Model selection and evaluation relied on precision, recall, F1, and ROC-AUC rather than raw accuracy, for the reasons demonstrated in the naive-baseline check below.

Naive Baseline

Before trusting any model's accuracy figure, we confirmed it beats naive guessing. A majority-class predictor already reaches 63.2% "accuracy" on this split while identifying zero malignant cases (0% precision, recall, and F1 on the positive class), a direct illustration of why accuracy alone is an unsafe metric on this dataset. A random (class-proportional) predictor reached 59.7% accuracy with 40.5% recall. Any retained model has to clear this floor by a wide margin, especially on recall.

Baseline Performance: Default Hyperparameters

Logistic Regression was trained with default hyperparameters (C = 1.0, L2 penalty) inside a pipeline with StandardScaler, fit only on the training fold to avoid leaking test-set statistics into scaling. On the held-out test set:

Metric	Value
Accuracy	97.37%
Precision (malignant)	100.0%
Recall (malignant)	92.86%
F1 (malignant)	96.30%
ROC-AUC	0.9940

5-Fold Cross-Validation

We used k = 5 rather than k = 10: with only 455 training rows, 10 folds would leave ~45 rows per fold, which is thin for a stable validation estimate, while 5 folds (~91 rows each) is a better bias/variance trade-off for a dataset this size. Cross-validating the full pipeline (scaler + model) ensures scaling is refit within each fold, avoiding leakage across folds.

Metric	Mean	Std. Dev.
Accuracy	97.58%	2.24%
ROC-AUC	0.9937	0.0069
F1	0.9665	0.0309

The fold-to-fold accuracy standard deviation (~2.2%) is small relative to the mean (~97.6%), indicating the model's performance is not highly sensitive to which 20% of the training data it happens to see — a reasonable basis for trusting the mean CV score as a stable estimate.

Hyperparameter Tuning

Three hyperparameters were tuned via GridSearchCV (5-fold CV, ROC-AUC scoring):

Parameter	Role	Why it affects performance
C	Inverse regularization strength (C = 1/λ)	Smaller C shrinks coefficients toward zero (less overfitting, more underfitting risk); larger C lets the model fit training data more closely, risking overfitting on a dataset this small.
penalty	Which norm regularizes the coefficients (L1 vs. L2)	L1 can zero out weak/redundant features (implicit feature selection); L2 shrinks all coefficients smoothly. With engineered and correlated features present, either could help, which is why both were searched.
solver	Optimization algorithm used to fit coefficients	liblinear supports both L1 and L2 penalties and is efficient at this dataset size, keeping the search compatible with both penalty options.

The best combination found was C = 10, L2 penalty, liblinear solver (best CV ROC-AUC = 0.9950). The table below compares the untuned baseline to the tuned model on the same held-out test set:

Metric	Baseline (default)	Tuned (C=10, L2)
Accuracy	97.37%	97.37%
Precision (malignant)	100.0%	97.56%
Recall (malignant)	92.86%	95.24%
F1 (malignant)	96.30%	96.39%
ROC-AUC (test)	0.9940	0.9868
ROC-AUC (5-fold CV)	0.9937	0.9950

Tuning left test-set accuracy essentially unchanged but shifted the error profile in a clinically favorable direction: recall on the malignant class rose from 92.9% to 95.2% (one additional true malignant case caught) at the cost of one extra false positive (precision fell from 100% to 97.6%). In a diagnostic context, trading a small amount of precision for higher recall is usually the right call, since a missed malignancy is far costlier than an unnecessary follow-up test. The 5-fold CV numbers, which are less sensitive to the specific 20% test split, confirm the gain is real: CV ROC-AUC improved from 0.9937 to 0.9950 with tuning.

Evaluation Metrics

Accuracy alone is not a safe metric here: Section "Naive Baseline" above shows a model can reach ~63% accuracy while catching zero malignant cases by exploiting the class imbalance. Because a false negative (a missed malignancy) is far costlier than a false positive (an unnecessary follow-up), we report recall/sensitivity and F1 alongside precision, with ROC-AUC as a threshold-independent summary.

Confusion matrix layout and formulas (positive class = malignant):

- Precision = TP / (TP + FP), of the cases flagged malignant, how many really were.
- Recall (Sensitivity) = TP / (TP + FN), of the cases that really were malignant, how many were caught. This is the metric we are most protective of.
- F1 = 2 × (Precision × Recall) / (Precision + Recall), balances both into a single number.

On the test set (113 cases), the tuned model produced the following confusion matrix:

	Predicted Benign	Predicted Malignant
Actual Benign	71 (TN)	1 (FP)
Actual Malignant	2 (FN)	40 (TP)

Critical Analysis

Limitations specific to this problem

- Logistic regression assumes a linear decision boundary in (scaled) feature space. The engineered ratio/interaction features help, but any non-linear interaction not captured by those four terms is invisible to the model.
- It is sensitive to multicollinearity. Milestone 2 already dropped features with pairwise correlation above 0.95, but some correlation remains among survivors, which can inflate coefficient variance (though it did not noticeably hurt accuracy here).
- With 455 training rows and 27 features, the model has relatively little data per parameter, part of why regularization (C) is tuned rather than left unconstrained.

Logistic Regression vs. Random Forest (alternative considered)

	Logistic Regression	**Random Forest**
Decision boundary	Linear (in scaled feature space)	Non-linear; captures feature interactions automatically
Interpretability	High — coefficients map directly to log-odds	Lower — requires feature importance / SHAP for similar insight
Overfitting risk here	Lower, especially with regularization	Higher without tuning depth/leaf size, though bagging offsets some of this
Training/inference cost	Very fast (single-digit ms)	Slower to train (many trees); still fast to predict
Feature scaling required	Yes	No

Preliminary comparisons against other model families (Random Forest, Gradient Boosting, and ensemble methods) suggest they edge out logistic regression on raw accuracy, largely because they can exploit non-linear feature interactions the linear model cannot. This is consistent with the project's original hypothesis — but the same rigor applied to the baseline here (formal 5-fold CV, hyperparameter tuning, and held-out evaluation) has not yet been applied to those models, so a confident comparison is reserved for the next milestone.

Proposed improvements for the next milestone

- Widen the C search (e.g., a log-spaced range from 1e-4 to 1e3) and add the saga solver to enable an elastic-net search over l1_ratio rather than choosing purely between L1 and L2.
- Use L1-selected coefficients, if L1 wins under a wider search, to explicitly drop near-zero-weight features and test whether a smaller feature set holds up under cross-validation.
- Apply the same rigor (CV, tuning, held-out evaluation) to a non-linear alternative (RBF-kernel SVM, Gradient Boosting) as the next milestone's point of comparison, since a few borderline error cases (below) hint that a non-linear boundary could resolve some remaining misclassifications.
- Use nested cross-validation for hyperparameter search so the reported CV score is not optimistically biased by using the same folds for both selection and evaluation.

Learning Curves, Error Analysis, and Reproducibility

A learning curve (training vs. validation accuracy across increasing training-set sizes) showed the two curves converging to a small gap as training size grows, with both plateauing at a high accuracy — the signature of a well-fit model rather than an overfit one (a persistent large gap would indicate overfitting; both curves plateauing low would indicate underfitting).

Error analysis on the 3 misclassified test cases (of 113) found two confidently-wrong predictions — malignant cases the model was confident were benign (predicted probabilities of 0.047 and 0.0001) — which is the more clinically concerning error type, since it would not be flagged for a second look. The third error was a benign case the model leaned malignant on (probability 0.755), a false alarm rather than a missed diagnosis. With only 3 errors, this sample is too small to claim a strong systematic pattern, but it is a lead worth following in the next milestone: checking whether the confidently-wrong malignant cases share unusually small size/shape-irregularity values relative to other malignant cases.

Both the baseline and tuned models fit and predicted in a few milliseconds, so training/inference time is not a practical constraint for this dataset size; model choice can be made purely on predictive performance and interpretability. All results are reproducible with random_state = 42 used throughout (train/test split, cross-validation folds, model initialization, and grid search).

Baseline Summary

Model	Accuracy	Precision	Recall	F1
Naive (Majority Class)	63.16%	0.00%	0.00%	0.00%
Naive (Random (stratified))	59.65%	44.74%	40.48%	42.50%
Baseline Logistic Regression (default)	97.37%	100.0%	92.86%	96.30%
Tuned Logistic Regression	97.37%	97.56%	95.24%	96.39%

The tuned logistic-regression baseline reaches ~97.4% test accuracy and 0.987 test ROC-AUC (0.995 under cross-validation), decisively beating both naive baselines and confirming the model is learning genuine signal from the 27-feature set rather than exploiting the mild class imbalance. This is the benchmark every subsequent model, the Random Forest comparison introduced above, and the additional ensemble methods (Gradient Boosting, AdaBoost, Stacking) explored alongside this milestone, must beat, with the same level of rigor, to justify its added complexity.